Introduction to Networking and the OSI Model Theory

1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Steps:-

1. Download and install **Wireshark** from the official website: **Wireshark Download**.
2. Open Wireshark and select your active **Wi-Fi or Ethernet** network interface.
3. Click **Start Capture**.
4. Open your browser and visit instagram.com.
5. Return to Wireshark and click the red **Stop** button.
6. Check the **Protocol** column in the packet list and note the protocols displayed.

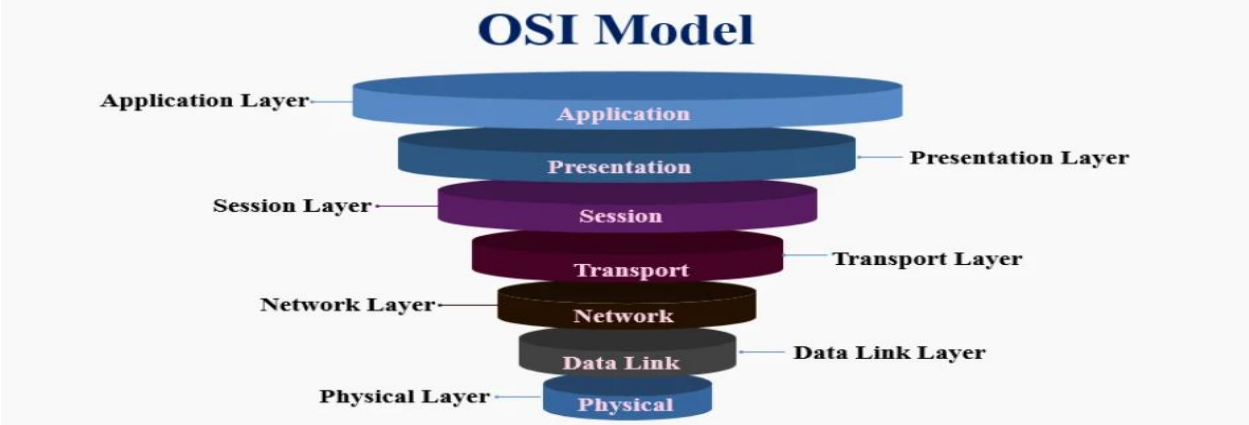
No.	Protocol	Function
1	DNS	Finds the IP address of a website.
2	TCP	Provides reliable data transmission.
3	TLSv1.2 / TLSv1.3	Helps encrypt website communication.
4	QUIC	Supports fast, secure internet communication.
5	UDP	Sends data packets without establishing a TCP connection.
6	HTTP	Transfers web data; HTTPS traffic is generally encrypted.
7	ICMP	Used for network diagnostic messages, when present.

2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

No.	Protocol	OSI Layer	Function
1	DNS	Layer 7 – Application	Finds the IP address of a website.
2	TCP	Layer 4 – Transport	Provides reliable data transmission.
3	IP	Layer 3 – Network	Routes packets from source to destination.

3. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



Layer	Name	Daily-Life Example
7	**Application**	WhatsApp — sending messages
6	**Presentation**	Instagram — displaying images and videos; encryption and data formats
5	**Session**	WhatsApp — maintaining a communication session
4	**Transport**	Video calls — delivering data using TCP or UDP
3	**Network**	Google Maps — IP routing helps data reach its destination
2	**Data Link**	Wi-Fi — transferring data between devices on a local network
1	**Physical**	Wi-Fi radio signals or Ethernet cables — carrying bits

4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.

How a WhatsApp Message Travels Through the OSI Model:-

Layer	Name	What Happens
7	**Application**	I type “Hello” and press Send in WhatsApp.
6	**Presentation**	The message is represented in a suitable data format and protected using encryption.
4	**Transport**	Transport protocols help deliver data between devices. WhatsApp calls and messages may use different protocols.
3	**Network**	IP addresses help route data across networks to its destination.
2	**Data Link**	Wi-Fi or mobile network technology carries data across the local network connection.
1	**Physical**	Radio signals or electrical signals carry the data as bits.